Decentralised Consensus for Indoor Multi-Robot Target Search and Tracking

Group 6

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1 Research Question

In a partially observable indoor simulation, does decentralised consensus-based coordination improve moving object search efficiency and localisation accuracy compared with centralised and communication-free approaches, particularly under robot failures and communication disturbances?

2 Motivation

Multi-robot teams can improve target localisation and tracking by combining observations from different viewpoints and coordinating exploration, as demonstrated by Co-NavGPT Yu et al. [2025]. Prior work has examined decentralised exploration under communication constraints Bayer and Faigl [2026], Matignon et al. [2012], but how decentralised consensus compares with centralised fusion and communication-free approaches under partial observability and communication failures remains relatively unexplored. This project investigates these trade-offs through a controlled indoor simulation, measuring localisation time, tracking accuracy, and robustness to faults, helping designers decide when the overhead of decentralised coordination is justified for resource-constrained robot teams.

3 Proposed Experiment

We propose a Python-based, two-dimensional indoor simulation with a predefined floor plan, obstacles, and initially random robot and target positions. Three mobile robots (baseline configuration, varied for scalability analysis) will explore and search for designated object(s) whose position changes at random intervals. Each robot will receive noisy, limited-range observations and maintain a local target-position estimate. On a user query, the system selects a robot to dispatch by jointly considering expected travel distance and the confidence (covariance) of its current target-position estimate, favouring a more distant but more certain robot over a nearer but less certain one; this selection is performed via a one-off query to each robot's current estimate at query time, independent of whichever coordination strategy governs the robots' ongoing tracking. To isolate the effect of estimation accuracy from transit-time re-planning, target relocation is paused for the duration of an active dispatch, resuming once the robot reaches its destination or the trial ends. We will compare three coordination strategies: (i) **Independent baseline**: robots search and estimate target positions without communication; (ii) **Centralised baseline**: robots transmit observations to a central coordinator, which applies the same innovation-based outlier detection as the proposed method to distinguish measurement noise from genuine target relocation, fuses observations accordingly, and broadcasts the updated target-position estimate to all robots; and (iii) **Proposed decentralised method**: robots exchange estimates with their communication neighbours and apply a consensus-based update rule that distinguishes measurement noise from genuine target relocation via innovation-based outlier detection, triggering a belief reset rather than blended averaging when relocation is detected. All three approaches will use identical sensing, robot dynamics, local navigation algorithms, and estimation logic, differing only in the communication topology through which target-position information is shared. In addition to this primary comparison, we will sweep the innovation-gating

threshold that governs the noise-versus-relocation decision, quantifying the resulting trade-off between false resets (noise misclassified as relocation) and missed relocations (genuine movement smoothed away as noise).

All methods will be evaluated using the same scenarios, with results compared through mean performance, confidence intervals, and graphical analysis.

- **Primary Metrics**
  - **Search efficiency**: Simulation time until the estimated target position is within a predefined distance tolerance of ground truth (reported with the success rate before a fixed time limit).
  - **Localisation and Tracking Accuracy**: Root-mean-square error between the estimated and ground-truth target positions over time.
- **Secondary Metrics**
  - **Scalability**: Changes in search efficiency and tracking accuracy as the number of robots increases, while keeping the environment and task conditions consistent.
  - **Robustness**: Degradation of search time and tracking error under robot failure or message loss, relative to fault-free trials.
  - **Communication cost**: Messages transmitted per robot per unit time across the three strategies, quantifying the coordination overhead each approach requires.
  - **Dispatch efficiency**: Rate of wasted dispatches (robot arrives but the target has since relocated or was never there) under uncertainty-aware versus nearest-estimate-only selection.
  - **Gating sensitivity**: False-reset rate and missed-relocation rate as a function of the innovation-gating threshold.

Possible Extension

Time permitting, we will investigate an adversarial fault scenario in which a compromised robot deliberately fabricates a large innovation to falsely trigger a belief reset in its neighbours, rather than merely dropping messages or injecting random noise. This targets a vulnerability specific to the proposed gating mechanism, distinguishing it from generic Byzantine fault-tolerance evaluations that assume only consistently false, rather than spoofed-event, reports.

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4 Expected Contribution

We hypothesise that decentralised consensus will outperform independent search while approaching centralised accuracy when communication is reliable, and that its innovation-based gating will allow it to track target relocations without the estimation error that naive averaging would introduce when robots disagree due to genuine movement rather than noise. We expect its response to failures to depend on communication-graph connectivity; the experiments will quantify when the improved information sharing offsets its communication and coordination costs. We further hypothesise that uncertainty-aware

dispatch will reduce wasted trips relative to nearest-estimate-only selection, and that the innovation-gating threshold will expose a measurable trade-off between false resets and missed relocations, informing how this parameter should be tuned in practice.

References

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